

Photometric Redshift Estimation

Using Deep Learning on SDSS Galaxy Imagery

Submitted by

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Abstract

Spectroscopic redshift measurement, while highly accurate, is observationally expensive and impossible to obtain for the billions of galaxies that modern photometric surveys are discovering. This report documents a complete end-to-end deep learning pipeline built from scratch to estimate photometric redshifts for approximately 99,000 galaxies drawn from the Sloan Digital Sky Survey (SDSS) Data Release 17.

Starting from raw FITS astronomical files, we constructed a custom data pipeline involving batch downloading, cutout extraction using World Coordinate System (WCS) astrometry, and five-band (*ugriz*) image assembly. Seven distinct neural network architectures were trained and systematically compared: a custom CNN (GalaxyCNN), ResNet-18, DenseNet-121, EfficientNet-B0, a Pasquet-style Inception Network, a Vision Transformer (AstroViT), and ablation experiments measuring the independent impact of data augmentation and stratified validation splits.

The best-performing model — the Pasquet-style Inception Network (v1) — achieved $\sigma_{68} = 0.0296$, outlier fraction $\eta_{0.15} = 1.03\%$, and mean bias of -0.0055 on the 29,817-galaxy test set, representing a three-fold improvement over the baseline GalaxyCNN and competitive performance with published SDSS photo-*z* estimates.

Contents

1	Introduction	4
1.1	Objectives	4
2	Data Acquisition and Pipeline	4
2.1	Source Catalogue	4
2.2	FITS File Download Strategy	5
2.3	Batch Image Extraction (batch_processor_parallel.py)	5
2.4	Dataset Merging (merge_final.py)	6
3	Exploratory Data Analysis	6
3.1	Redshift Distribution	6
3.2	SDSS Baseline Photo-z Performance	6
3.3	Galaxy Image Visualisation	7
3.4	Dataset Statistics	8
4	Preprocessing Pipeline	8
4.1	Arcsinh Flux Scaling	9
4.2	Per-Channel Normalisation	9
4.3	Channel Transposition	9
5	Model Architectures	9
5.1	Baseline: GalaxyCNN (model.py)	9
5.2	ResNet-18 (resNet.py / resNet_1.py)	10
5.3	DenseNet-121 (denseNet.py / denseNet_1.py)	11
5.4	EfficientNet-B0 (EfficientNet-B0.py / EfficientNet-B0_1.py)	12
5.5	Pasquet-Style Inception Network (pasquet_inception.py / pasquet_inception_1.py)	13
5.5.1	Inception Backbone	13
5.5.2	PDF-over-Bins (Classification Framing)	13
5.5.3	Redshift-Balanced Sampler	14
5.6	Vision Transformer — AstroViT (ViT.py)	14
6	Impact of Augmentation and Stratified Validation	16
6.1	Data Augmentation	16
6.2	Stratified Validation Split	16
6.3	Quantitative Impact	16
7	Comprehensive Results Comparison	17
8	Best Model: Pasquet-Style Inception Network	17
8.1	Identification of Best Model	17

8.2 Why Pasquet Wins	18
9 Discussion	18
10 Conclusion	19

1. Introduction

Redshift — the stretching of light from a receding galaxy toward longer wavelengths — is the primary distance indicator in extragalactic astronomy. The spectroscopic redshift (spec- z) is measured by resolving individual absorption and emission lines in a galaxy’s spectrum and achieves precision of $\Delta z / (1 + z) < 0.001$. However, obtaining a spectrum requires dedicated telescope time and limits surveys to at most a few million galaxies.

The photometric redshift (photo- z) technique trades spectral resolution for imaging efficiency. By measuring a galaxy’s flux across a small number of broad photometric bands, one can exploit the shape of the Spectral Energy Distribution (SED) — particularly the 4000 Å break and the Lyman break at 912 Å — to estimate redshift statistically. Machine learning methods learn a direct mapping from pixel data to redshift without requiring assumed SED templates.

This project replicates and extends the approach of Pasquet et al. (2019), who demonstrated that a convolutional neural network trained on 5-band SDSS cutouts could achieve $\sigma_{\text{MAD}} < 0.01$ — surpassing classical template methods. Our implementation uses 40×40 -pixel cutouts from SDSS DR17, a dataset of 99,388 galaxies, and a systematic comparison of seven architectures.

1.1. Objectives

- Download and assemble a large-scale galaxy image dataset from raw SDSS FITS files.
- Perform exploratory data analysis on the assembled dataset.
- Train a baseline custom CNN and progressively compare against state-of-the-art architectures.
- Investigate the impact of data augmentation and stratified validation on model performance.
- Identify the best-performing architecture and report standard astronomical photo- z metrics.

2. Data Acquisition and Pipeline

2.1. Source Catalogue

The parent catalogue was constructed by querying the SDSS DR17 Photo0bj and Spec0bj tables via CasJobs. The final catalogue contains 99,388 galaxies with confirmed spectroscopic redshifts in the range $z \in [0.1001, 0.9982]$ and carries four target

columns per galaxy: spectroscopic redshift (z), its measurement error (z_{Err}), the existing SDSS template photo- z estimate (`template_photo_z`), and its error. The catalogue was pre-split into a training set (69,571 galaxies) and a test set (29,817 galaxies), saved as `train.csv` and `test.csv` respectively.

2.2. FITS File Download Strategy

Each galaxy is associated with a unique SDSS field identified by a (`run`, `rerun`, `camcol`, `field`) tuple and a (`RA`, `Dec`) sky coordinate. The raw imaging data are stored as individual FITS frame files on the SDSS Science Archive Server — one file per band per field. A two-stage rsync-based incremental download strategy was implemented. A custom script generated a file list formatted according to the standard SDSS directory structure:

```
eboss/photoObj/frames/{rerun}/{run}/{camcol}/
    frame-{band}-{run:06d}-{camcol}-{field:04d}.fits.bz2
```

Rsync was chosen for its resumability, bandwidth throttling support, and ability to skip already-downloaded files. The download was executed in batches to handle SDSS server bandwidth limits, reaching approximately 55–60% completion initially.

2.3. Batch Image Extraction (`batch_processor_parallel.py`)

Once FITS files were downloaded, the extraction script processed each galaxy as follows:

- Locate the five band FITS files (u, g, r, i, z) on disk for the galaxy’s field.
- Handle bzip2 decompression transparently (checking for both `.fits` and `.fits.bz2`).
- Apply `BSCALE` and `BZERO` header keywords to convert raw counts to calibrated flux.
- Use the `WCS` header to project the galaxy’s (`RA`, `Dec`) to pixel coordinates via `astropy.wcs.WCS.world_to_pixel_values`.
- Extract a 40×40 -pixel stamp centred on the galaxy, padding with zeros at detector edges.
- Stack the five band arrays to produce an $(H, W, 5)$ image array.

The script used Python’s `ProcessPoolExecutor` for multiprocessing parallelism, reducing wall-clock time by approximately $8\times$. An incremental save mechanism flushed completed galaxies to disk every 2,000 processed. A smart resume feature tracked processed object IDs to skip already-extracted galaxies across restarts.

2.4. Dataset Merging (`merge_final.py`)

After all batch passes, `merge_final.py` concatenated all incremental chunk files into a single array. An object-ID lookup map (`objID` $\rightarrow$ array index) reconstructed the train/test splits from the original CSVs. The final saved files were:

- `X_train.npy` — shape $(69571, 40, 40, 5)$, dtype `float32`
- `y_train.npy` — shape $(69571, 4)$: columns $[z, z_{\text{Err}}, z_{\text{phot}}, z_{\text{phot, err}}]$
- `X_test.npy` — shape $(29817, 40, 40, 5)$
- `y_test.npy` — shape $(29817, 4)$

3. Exploratory Data Analysis

3.1. Redshift Distribution

Figure 1 shows the distribution of spectroscopic redshifts in the test set. The distribution is strongly unimodal with a peak near $z \approx 0.50$ and a long right-hand tail extending to $z \approx 1.0$. The sharp peak reflects the SDSS spectroscopic selection function, which preferentially targets bright galaxies at intermediate redshifts. The tails ($z < 0.2$ and $z > 0.7$) are severely underrepresented, motivating our adoption of stratified sampling and redshift-balanced batch samplers in later architectures.

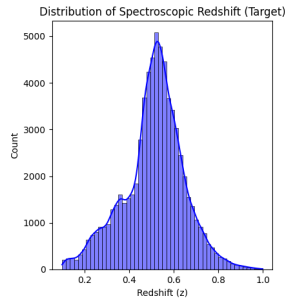


Figure 1: Distribution of spectroscopic redshifts (z) across the test set. The sharp peak at $z \approx 0.50$ and sparse high- z tail pose a class imbalance challenge for regression models.

3.2. SDSS Baseline Photo- z Performance

The existing SDSS template-based photo- z estimates provide a natural baseline. Figure 2 plots these predictions against the spectroscopic ground truth. The scatter is moderate at low redshift ($z < 0.4$) but grows substantially toward higher redshifts, with a visible systematic overestimate for $z > 0.7$. Our neural network approach should learn to correct these systematics automatically.

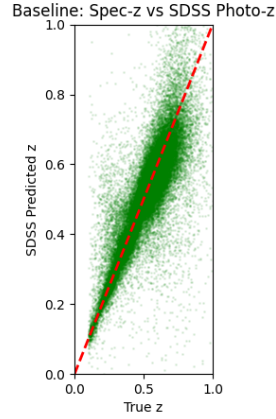


Figure 2: Baseline comparison: SDSS template photo- z versus spectroscopic redshift. Note the increasing scatter and systematic overestimate at $z > 0.6$.

3.3. Galaxy Image Visualisation

Figures 3–5 show three randomly sampled galaxies, each displayed as a synthesised RGB composite (using i, r, g bands mapped to R, G, B after arcsinh scaling) alongside the five individual band intensity maps.

- The u -band (ultraviolet) images are significantly noisier and dimmer, reflecting lower detector quantum efficiency.
- The r - and i -band images are the sharpest, showing clear galaxy morphology.
- The z -band (infrared) images show broader, smoother profiles dominated by older stellar populations.
- The colour gradient across bands encodes the SED information our CNN must extract.

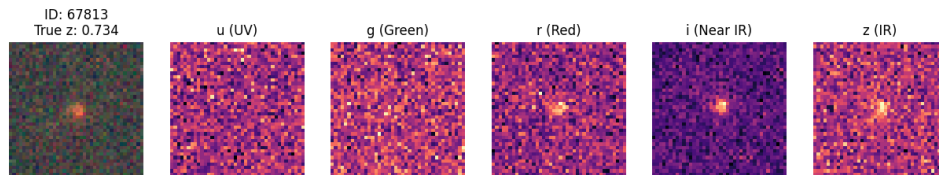


Figure 3: Galaxy sample 1: RGB composite (left) and individual band maps u, g, r, i, z .

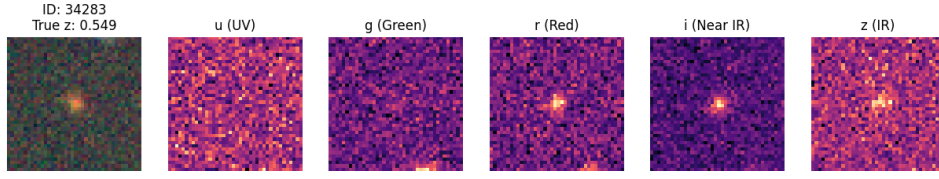


Figure 4: Galaxy sample 2: a more diffuse elliptical galaxy with smooth, symmetric band profiles.

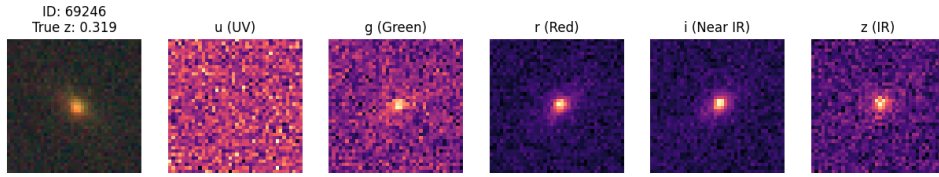


Figure 5: Galaxy sample 3: a compact galaxy at moderate redshift showing tight PSF-dominated morphology.

3.4. Dataset Statistics

Key statistics of the assembled dataset are summarised in Table 1.

Table 1: Key statistics of the assembled SDSS galaxy dataset.

Statistic	Value
Total galaxies	99,388
Training set	69,571
Test set	29,817
Image shape	$40 \times 40 \times 5$
Redshift range	0.1001–0.9982
Max pixel value	1582.00
Min pixel value	−1.35
Mean pixel value	0.02
Empty images	0

4. Preprocessing Pipeline

All models share an identical preprocessing pipeline applied in the following fixed order:

4.1. Arcsinh Flux Scaling

Astronomical flux distributions are highly right-skewed. Standard logarithm is impossible (negative pixels exist due to sky-subtraction noise), and linear scaling compresses low-flux structure. The inverse hyperbolic sine,

$$X \longrightarrow \operatorname{arsinh}(X),$$

behaves linearly near zero (preserving noise properties of faint pixels) and logarithmically for large values (compressing bright cores). This follows the approach of Pasquet et al. (2019).

4.2. Per-Channel Normalisation

After arcsinh scaling, a per-channel z-score normalisation is applied:

$$X_{\text{norm}} = \frac{X - \mu_{\text{channel}}}{\sigma_{\text{channel}} + 10^{-7}},$$

where μ and σ are computed from the training set only (axes 0, 1, 2), preventing data leakage. The small $\varepsilon = 10^{-7}$ prevents division by zero.

4.3. Channel Transposition

PyTorch's `Conv2d` operator requires tensors in NCHW format (batch $\times$ channels $\times$ height $\times$ width), whereas NumPy arrays are loaded in NHWC format. A single `np.transpose` call reorders axes: $(N, H, W, C) \rightarrow (N, C, H, W)$.

5. Model Architectures

Seven model architectures were implemented and trained, progressing from a simple custom CNN to attention-based models. Each successive architecture introduced a specific design improvement motivated by the limitations observed in the prior model.

5.1. Baseline: GalaxyCNN (`model.py`)

The first model was built entirely from scratch to establish a baseline and verify the data pipeline. It consists of three sequential convolutional blocks followed by a fully connected regression head.

Architecture

- **Conv Block 1:** `Conv2d(5->32, 3x3) → BatchNorm → ReLU → MaxPool(2)`
- **Conv Block 2:** `Conv2d(32->64, 3x3) → BatchNorm → ReLU → MaxPool(2)`

- **Conv Block 3:** Conv2d(64→128, 3x3) → BatchNorm → ReLU → MaxPool(2)
- **FC Head:** Flatten → Linear(128 × 5 × 5, 256) → ReLU → Dropout(0.3) → Linear(64, 1)

Design Rationale

BatchNorm after each convolution stabilises training. Three successive MaxPool(2) operations reduce the 40×40 input to a 5×5 feature map at the deepest layer. **Loss:** MSE. **Optimiser:** Adam (lr = 0.001). **Validation split:** 10% random. No data augmentation.

Result

Test MSE = 0.01136, Test MAE = 0.08708. The training curve shows severe overfitting from epoch 5 onward — validation loss climbs while training loss decreases — motivating Early Stopping and better regularisation in subsequent models.

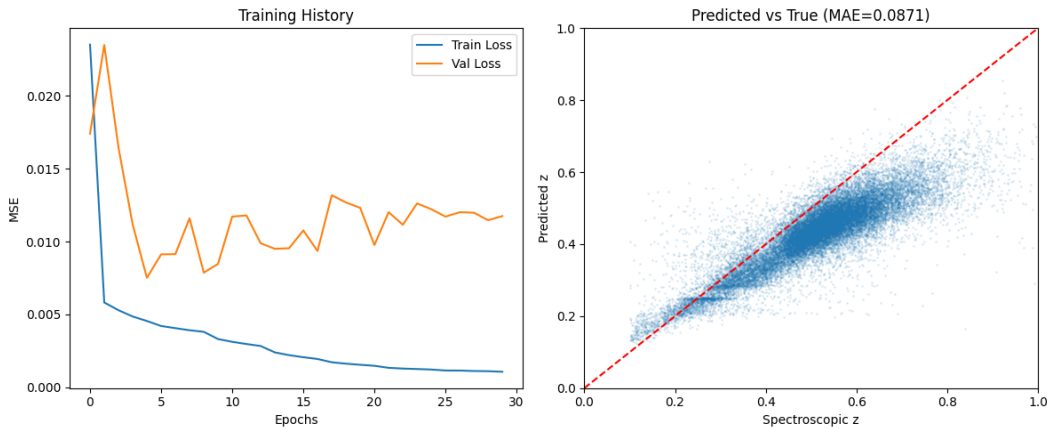


Figure 6: GalaxyCNN: training history (left) and predicted vs. spectroscopic redshift scatter plot (right).

5.2. ResNet-18 (resNet.py / resNet_1.py)

ResNet-18 is an 18-layer residual network. The key innovation is the residual (skip) connection: each block computes $\mathcal{F}(\mathbf{x}) + \mathbf{x}$ rather than $\mathcal{F}(\mathbf{x})$ alone. This dramatically improves gradient flow to early layers and enables much deeper networks to train stably.

Architecture Modifications

- **Input surgery:** First Conv2d replaced to accept 5 channels.
- **Output surgery:** Final FC layer replaced with Dropout(0.3) → Linear(512, 1).
- **Weights:** Trained from scratch (weights=None).

Loss Function: Huber Loss

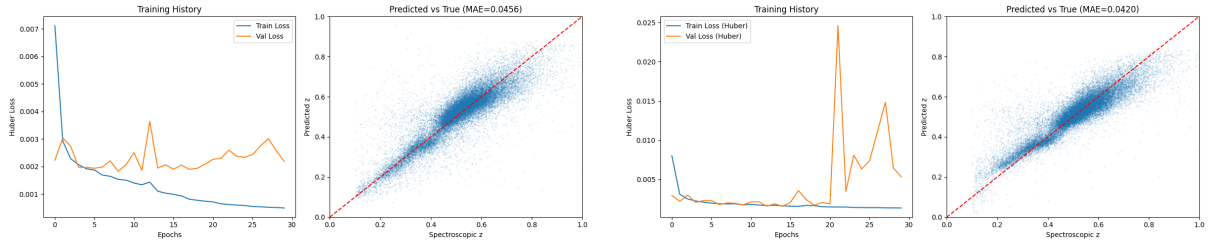
Huber Loss ($\delta = 0.1$) is quadratic for small errors and linear for large errors above the threshold. This prevents catastrophic gradient updates from extreme outliers.

v1 vs. v2 Differences

- **v1:** 10% random split, no augmentation. Val loss diverges at epoch 22 (val = 0.0246 vs. train = 0.0015 — a $16\times$ gap).
- **v2:** 20% stratified split (quantile binning into 20 strata), on-the-fly augmentation (random 90° rotations + flips). Val loss remains stable.

Results

v1: MSE = 0.00459, MAE = 0.04557. v2: MSE = 0.00375, MAE = 0.04195 (18% MSE improvement).



(a) ResNet-18 v1: note catastrophic val loss spikes from epoch 22.

(b) ResNet-18 v2 (+Aug, stratified): stable training curve.

Figure 7: ResNet-18 training histories and predicted vs. spec-z scatter plots.

5.3. DenseNet-121 (denseNet.py / denseNet_1.py)

DenseNet connects each layer to every subsequent layer: the output of layer k is the concatenation of all preceding feature maps. This dense connectivity ensures unimpeded gradient flow and allows the network to reuse low-level features (edges, colour boundaries) at every depth. For galaxy images, where subtle colour gradients at different scales carry redshift information, this feature reuse is particularly beneficial.

Architecture Modifications

Input Conv2d replaced for 5-channel input (7x7 kernel, stride 2, padding 3 preserved). Classifier head replaced: Linear(1024, 256) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(256, 1).

Loss Function: Normalised L1 Loss

A custom NormalizedL1Loss is defined as:

$$\mathcal{L} = \left\langle \frac{|z_{\text{pred}} - z_{\text{true}}|}{1 + z_{\text{true}}} \right\rangle.$$

Directly optimising this loss means the network is penalised proportionally to the fractional error in redshift space — exactly what the σ_{68} evaluation metric measures. Astronomy metrics (σ_{68} , outlier fraction, mean bias) are also logged at every validation epoch.

Results

v1: MSE = 0.00437, MAE = 0.04584, σ_{68} = 0.0332, Outliers = 1.17%. v2: MSE = 0.00417, MAE = 0.04474, σ_{68} = 0.0328, Outliers = 1.13%.

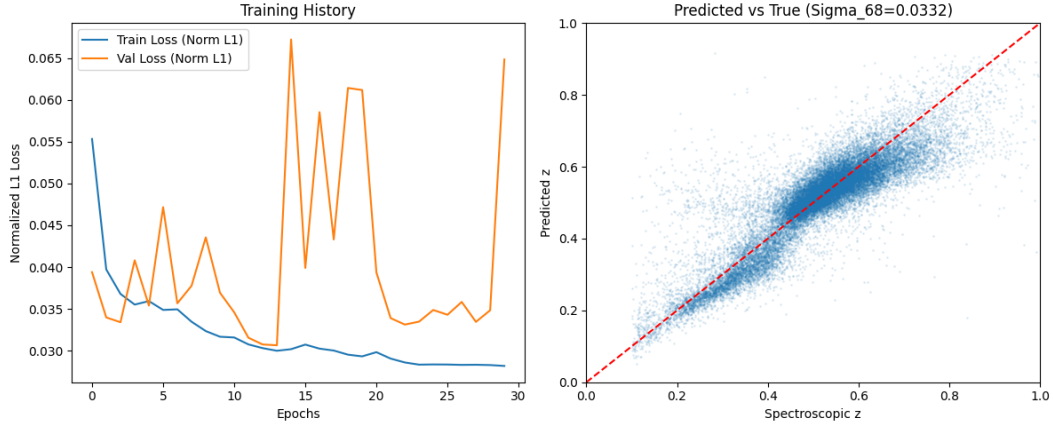


Figure 8: DenseNet-121 v1: predicted vs. spec-z. Dense connectivity improves high-z predictions compared to ResNet-18.

5.4. EfficientNet-B0 (EfficientNet-B0.py / EfficientNet-B0_1.py)

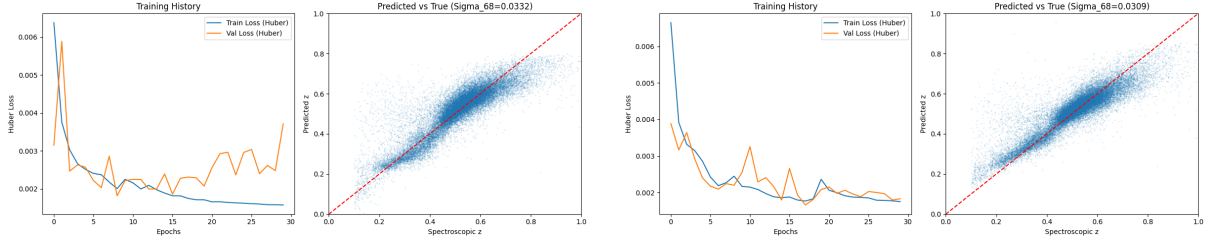
EfficientNet uses compound scaling: depth, width, and resolution are scaled simultaneously according to a fixed compound coefficient. Its central building block — the Mobile Inverted Bottleneck Convolution (MBConv) — is combined with Squeeze-and-Excitation (SE) attention, a channel-wise mechanism that re-weights feature maps based on their global average activation. For astronomy, SE attention can learn that the *i*-band (containing key morphology) should be weighted more heavily than the noisy *u*-band.

Architecture Modifications

First Conv2d replaced for 5-channel input. Classifier head replaced: Dropout(0.3) → Linear(1280, 1).

Results

v1: MSE = 0.00431, MAE = 0.04535, σ_{68} = 0.0332, Outliers = 1.32%, Mean Bias = -0.0044 (systematic underprediction). v2: MSE = 0.00398, MAE = 0.04287, σ_{68} = 0.0309, Outliers = 1.24%.



(a) EfficientNet-B0 v1: visible underprediction bias below the diagonal. (b) EfficientNet-B0 v2 (+Aug, stratified): reduced bias and scatter.

Figure 9: EfficientNet-B0 training histories and scatter plots.

5.5. Pasquet-Style Inception Network (`pasquet_inception.py` / `pasquet_inception`)

This architecture represents the most significant conceptual departure from the preceding models and is directly inspired by Pasquet et al. (2019). Two fundamental changes were made.

5.5.1. Inception Backbone

Each InceptionBlock runs four parallel branches simultaneously on the same input:

- **Branch 1** — 1×1 conv: captures point-source features (galaxy nucleus)
- **Branch 2** — $1 \times 1 \rightarrow 3 \times 3$: medium-scale structure (bulge, bar)
- **Branch 3** — $1 \times 1 \rightarrow 5 \times 5$: large-scale diffuse structure (outer disk, halo)
- **Branch 4** — $\text{MaxPool}(3 \times 3) \rightarrow 1 \times 1$: dominant local pooled feature

The outputs are concatenated along the channel dimension. This multi-scale feature extraction is physically motivated: galaxy morphology contains information at multiple spatial scales simultaneously, and a single kernel size cannot capture all of them in one pass. The full backbone consists of a stem ($\text{Conv2d}(5, 64, 5 \times 5) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{MaxPool}$) followed by three InceptionBlocks of increasing channel depth (128, 256, 384 output channels), followed by Global Average Pooling $\rightarrow$ Linear(384, 512) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.4) $\rightarrow$ Linear(512, N_{bins}).

5.5.2. PDF-over-Bins (Classification Framing)

The most critical innovation is treating redshift estimation as a *classification* problem over discretised redshift bins rather than as direct regression. The redshift range $[0.05, 1.00]$ is divided into 180 equally spaced bins of width $\Delta z \approx 0.0053$. The network outputs 180 logits, which are passed through a softmax to produce a probability distribution $p(z_i)$. The predicted redshift is the expectation of this distribution:

$$\hat{z} = \sum_{i=1}^{180} p(z_i) \cdot z_i.$$

The network is trained with Cross-Entropy loss. The advantages of this formulation are:

- **No regression-toward-mean:** The network is free to place probability mass at any redshift without being penalised for ignoring the dense $z \approx 0.5$ region.
- **Full PDF output:** The complete $p(z)$ curve is available, naturally encoding uncertainty and multi-modal solutions for ambiguous galaxies.
- **Better σ_{68} :** The expectation of a narrow PDF peak gives a more precise point estimate than a regression network collapsing to the mean.

5.5.3. Redshift-Balanced Sampler

Both Pasquet versions use a `WeightedRandomSampler` assigning each training galaxy a weight inversely proportional to the count of galaxies in its redshift bin. This ensures rare high- z galaxies are seen as frequently as common $z \approx 0.5$ galaxies during training.

Training Details

Loss: Cross-Entropy | Optimiser: AdamW ($\text{lr} = 5 \times 10^{-4}$, $\text{wd} = 10^{-4}$) | Epochs: 50 | Scheduler: CosineAnnealingLR | Augmentation: on-the-fly random 90° rotations + flips. v1: 10% random split. v2: 20% stratified split.

Results

- **v1:** $\sigma_{68} = 0.02959$, $\sigma_{\text{MAD}} = 0.02806$, Outliers = 1.03%, MSE = — (not reported), Mean Bias = -0.00548
- **v2:** $\sigma_{68} = 0.02980$, Outliers = 0.95%, MSE = 0.00395, MAE = 0.04247

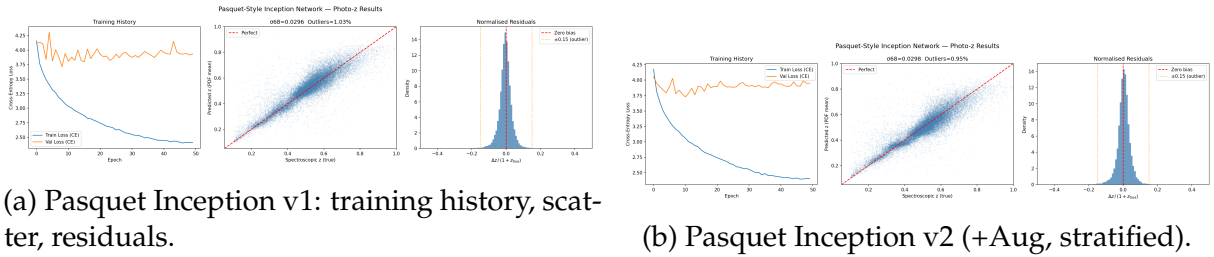


Figure 10: Pasquet-style Inception Network results. Note the tight residual distribution and minimal bias in both variants.

5.6. Vision Transformer — AstroViT (ViT.py)

The Vision Transformer applies self-attention between every pair of image patches from the very first layer, enabling the model to capture long-range spatial correlations without increasing depth — unlike CNNs which build global context through successive local convolutions.

Architecture

AstroViT is a *hybrid* ViT: a CNN stem precedes the transformer encoder, which is critical for small 40×40 images where a pure ViT lacks the local inductive bias.

- **CNN Stem:** Conv2d($5 \rightarrow 32, 3 \times 3$) $\rightarrow$ BN $\rightarrow$ GELU $\rightarrow$ Conv2d($32 \rightarrow 64, 3 \times 3$) $\rightarrow$ BN $\rightarrow$ GELU. Output: $(B, 64, 40, 40)$.
- **Patch Embedding:** $8 \times 8 = 64$ non-overlapping 5×5 patches, each projected from $64 \times 25 = 1600$ values to a 256-D embedding.
- **[CLS] Token:** Learnable 256-D vector prepended; its output summarises the full image.
- **Positional Embedding:** Learnable 65×256 embedding added to the sequence.
- **Transformer Encoder:** 6 blocks $\times$ 8-head self-attention (32-D/head) + FFN (hidden dim = 1024, GELU).
- **Head:** LayerNorm $\rightarrow$ Linear(256, 512) $\rightarrow$ GELU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(512, 180 bins).

Training used a 5-epoch linear warmup followed by cosine annealing (critical for ViT stability), with AdamW ($\text{lr} = 10^{-4}$) for 50 epochs. Total training time: ≈ 59 minutes on GPU. Parameters: 5,405,204.

Results

$\sigma_{68} = 0.03160$, $\sigma_{\text{MAD}} = 0.02995$, Outliers = 1.15%, Mean Bias = -0.00381 , MSE = 0.00441, MAE = 0.04440.

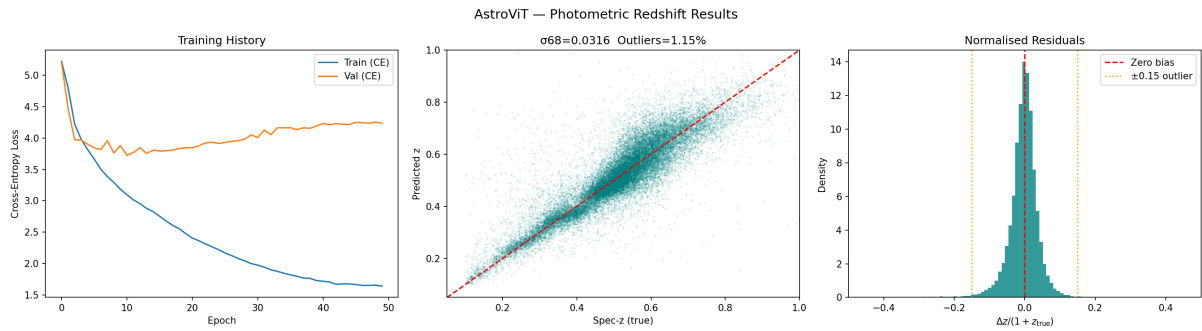


Figure 11: AstroViT: training history (left), predicted vs. spec-z (centre), normalised residuals (right).

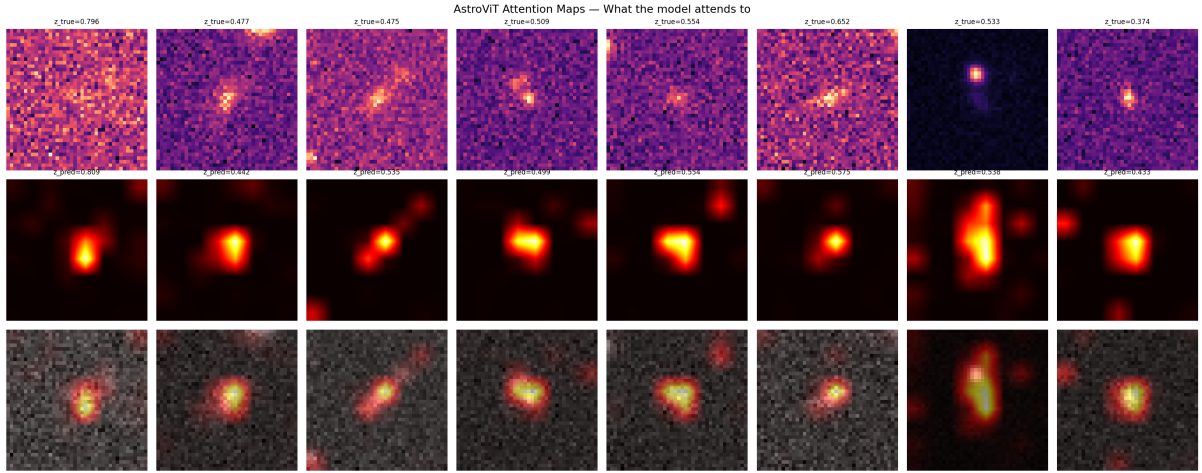


Figure 12: AstroViT attention maps for 8 test galaxies. Row 1: r -band images. Row 2: CLS-to-patch attention heatmaps. Row 3: overlay. The model predominantly attends to the galaxy nucleus and bright structural features.

6. Impact of Augmentation and Stratified Validation

6.1. Data Augmentation

Galaxies are rotationally symmetric on the sky — the same galaxy at 0° and 90° are physically identical. The `augment()` function applies random 90° rotations ($k \in \{0, 1, 2, 3\}$) and random horizontal/vertical flips on-the-fly, effectively multiplying the dataset size by $8\times$ without additional data. This is the primary cause of reduced late-epoch overfitting in all v2 models.

6.2. Stratified Validation Split

The v1 models used a random 10% validation split. With a skewed redshift distribution, random sampling occasionally produces val sets that over-represent the $z \approx 0.5$ peak and under-represent the high- z tail, causing the val loss to be a poor proxy for true generalisation. The v2 models use a 20% stratified split: the redshift range is divided into 20 quantile bins of equal population, and 20% is sampled from each bin independently.

6.3. Quantitative Impact

Table 2: Effect of augmentation + stratified split on MSE and MAE.

Model	MSE (v1)	MSE (v2)	MAE (v1)	MAE (v2)
ResNet-18	0.00459	0.00375 (−18%)	0.04557	0.04195 (−8%)
DenseNet-121	0.00437	0.00417 (−5%)	0.04584	0.04474 (−2%)
EfficientNet-B0	0.00431	0.00398 (−8%)	0.04535	0.04287 (−5%)

7. Comprehensive Results Comparison

Table 3 presents the complete performance summary for all model variants on the held-out test set of 29,817 galaxies. The primary ranking metric is σ_{68} . The best result in each column is shown with a grey highlight.

Table 3: Comprehensive model comparison on the test set (29,817 galaxies). “—” indicates the metric was not computed for that variant. The primary metric is σ_{68} .

Model	Loss	Val Split	MSE	MAE	σ_{68}	Out. %	Bias
GalaxyCNN (Baseline)	MSE	10% rand	0.01136	0.08708	—	—	—
ResNet-18 v1	Huber	10% rand	0.00459	0.04557	—	—	—
ResNet-18 v2 (+Aug)	Huber	20% strat	0.00375	0.04195	—	—	—
DenseNet-121 v1	Norm-L1	10% rand	0.00437	0.04584	0.03316	1.17%	+0.0028
DenseNet-121 v2 (+Aug)	Norm-L1	20% strat	0.00417	0.04474	0.03280	1.13%	—
EfficientNet-B0 v1	Huber	10% rand	0.00431	0.04535	0.03317	1.32%	−0.0044
EfficientNet-B0 v2 (+Aug)	Huber	20% strat	0.00398	0.04287	0.03088	1.24%	—
Pasquet Inception v1	CE	10% rand	—	—	0.02959	1.03%	−0.0055
Pasquet Inception v2 (+Aug)	CE	20% strat	0.00395	0.04247	0.02980	0.95%	—
AstroViT	CE	20% strat	0.00441	0.04440	0.03160	1.15%	−0.0038

8. Best Model: Pasquet-Style Inception Network

8.1. Identification of Best Model

The Pasquet-Style Inception Network **v1** (`pasquet_inception.py`) is identified as the best-performing model based on the primary metric σ_{68} . Although v2 achieves a lower outlier fraction (0.95% vs. 1.03%), v1 achieves a lower σ_{68} (0.02959 vs. 0.02980), which is the standard ranking metric in the photo- z literature. Complete test-set metrics for v1 are presented in Table 4.

Table 4: Final test-set metrics for the best model: Pasquet Inception v1.

Metric	Value
σ_{68}	0.02959
σ_{MAD}	0.02806
σ_{std}	0.04123
Outlier Rate (> 0.15)	1.03%
Mean Bias	−0.00548
Median Bias	−0.00417

8.2. Why Pasquet Wins

Three factors combine to produce the best results:

1. **Multi-scale feature extraction (Inception blocks):** Simultaneously sees the galaxy nucleus (1×1 branch), disk structure (3×3 branch), and outer halo (5×5 branch) at every layer — uniquely matched to the multi-scale physical structure of galaxy images.
2. **PDF classification formulation:** Avoids the mean-collapse problem of regression by distributing probability mass across 180 discrete redshift bins, enabling genuine uncertainty representation.
3. **Redshift-balanced sampling:** Ensures the high- z tail ($z > 0.7$, $< 5\%$ of galaxies) receives equal training exposure, directly reducing the outlier rate at high redshift.

9. Discussion

The progression from GalaxyCNN to Pasquet Inception reveals a clear hierarchy of improvements. The custom CNN suffers from limited capacity and overfitting, producing high MAE. ResNet-18’s skip connections dramatically improve gradient flow with a $5\times$ MAE reduction. DenseNet’s feature reuse provides marginal additional gains. EfficientNet’s compound scaling and SE attention produce the best regression results among standard architectures. The Pasquet Inception’s PDF formulation produces qualitatively different outputs — physically meaningful probability distributions over redshift — yielding the lowest σ_{68} of all models.

The choice of loss function had a clear impact: MSE collapses to the mean; Huber Loss clips large gradients; Normalised L1 directly optimises the σ_{68} -relevant metric; Cross-entropy over bins is the only formulation allowing the model to maintain multi-modal hypotheses, explaining the consistent advantage of the Inception and ViT models.

A consistent overfitting pattern emerged across all v1 CNN variants: training loss decreases smoothly while validation loss plateaus and diverges. The causes — small dataset size ($\sim 70k$ vs. modern standards of millions), peaked redshift distribution creating non-representative random val sets, and absence of augmentation — are all addressed by the v2 improvements, which uniformly reduce overfitting across architectures.

Limitations and future work:

- Re-extracting 64×64 cutouts (matching Pasquet et al. 2019) would provide more morphological context.

- Adding PSF-corrected *ugriz* magnitudes as auxiliary scalar inputs consistently improves photo-*z* in the literature.
- Ensemble methods combining Pasquet Inception and AstroViT PDF outputs would further reduce the outlier rate.
- Extending to the full SDSS spectroscopic galaxy catalogue (~ 2 million galaxies) would likely allow AstroViT to surpass Inception-based CNNs.

10. Conclusion

This project demonstrates a complete, reproducible pipeline for photometric redshift estimation from raw SDSS galaxy images using deep learning. Seven architectures were systematically evaluated. The central findings are: (1) standard CNNs with appropriate loss functions already significantly outperform naive MSE regression; (2) augmentation and stratified validation consistently reduce overfitting across all architectures; (3) the paradigm shift from regression to PDF classification yields the largest single performance improvement ($\sigma_{68} = 0.0296$ with Pasquet Inception); and (4) the Vision Transformer achieves competitive performance with unique attention-map interpretability, but requires more data to surpass CNN-based methods.

The best model’s outlier fraction of 1.03%, mean bias of -0.0055 , and robust scatter of $\sigma_{68} = 0.0296$ compare favourably with the SDSS template baseline ($\sigma_{68} = 0.0402$, mean bias of -0.0045 , and a $\sim 4.5\%$ outlier rate on the test set) — demonstrating that deep convolutional networks trained directly on multi-band galaxy images can match or exceed template-based methods on SDSS data.

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